

Homework 2

MSBA 400: Statistical Foundations for Data Analytics
Professor Rossi

HW 2 Solutions

Question 1

Q1, part A

In the class notes, we introduced the concept of R^2 . Show that the formula $R^2 = \frac{SSR}{SST}$ implies that R^2 is the square of the sample correlation coefficient between X and Y , r_{XY} . Hint: recall from the notes how the fitted regression line can be expressed in terms of deviations from the mean.

Fitted regression line, can be written as:

$$\hat{Y}_i = b_0 + b_1 X_i$$

Substituting for intercept (b_0) from LS formula and re-arrange

$$\hat{Y}_i - \bar{Y} = b_1(X_i - \bar{X})$$

$$\begin{aligned} R^2 &= \frac{SSR}{SST} = \frac{\sum_{i=1}^N (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= \frac{\sum_{i=1}^N (\hat{Y}_i - \bar{Y})(\hat{Y}_i - \bar{Y})}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \end{aligned}$$

using (1), we re-write RHS as

$$\begin{aligned} &= \frac{\sum_{i=1}^N b_1^2 (X_i - \bar{X})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= b_1^2 \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \end{aligned}$$

using the definition of b_1 in terms of r_{XY}

$$= r_{XY}^2 \times \frac{s_Y^2}{s_X^2} \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2}$$

using the definition of s_X and s_Y

$$\begin{aligned} &= r_{XY}^2 \times \frac{s_Y^2}{s_X^2} \frac{s_X^2}{s_Y^2} \\ &= r_{XY}^2 \end{aligned}$$

Q1, part B

In the class notes, we used intuition to argue the regression property, $\text{corr}(X, e) = 0$. Show this directly results from the formula for b_1 . Hint: substitute, $e_i = Y_i - b_0 - b_1 X_i$ into $\text{corr}(X, e)$.

From the LS formula we have

$$b_1 = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^N (X_i - \bar{X})^2}$$

$$\begin{aligned}
b_1 \sum_{i=1}^N (X_i - \bar{X})^2 &= \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y}) \\
&\text{using } Y_i = b_0 + b_1 X_i + e_i \\
&= \sum_{i=1}^N (X_i - \bar{X})(b_0 + b_1 X_i + e_i - \bar{Y}) \\
&= \sum_{i=1}^N (X_i - \bar{X})(b_0 + b_1 X_i - \bar{Y}) + \sum_{i=1}^N (X_i - \bar{X})(e_i) \\
&\text{substituting for intercept } (b_0) \text{ from LS formula} \\
&= \sum_{i=1}^N (X_i - \bar{X})(\bar{Y} - b_1 \bar{X} + b_1 X_i - \bar{Y}) + \sum_{i=1}^N (X_i - \bar{X})(e_i) \\
&= b_1 \sum_{i=1}^N (X_i - \bar{X})^2 + \sum_{i=1}^N (X_i - \bar{X})(e_i) \\
\sum_{i=1}^N (X_i - \bar{X})(e_i) &= 0 \implies \text{Cov}(X, e) = \text{Corr}(X, e) = 0
\end{aligned}$$

Question 2 : More on Nearest Neighbor Approaches

The simplest view of nearest neighbor methods is to “slice” into the data for only a small interval of X values. This distribution is called the conditional distribution of Y given X .

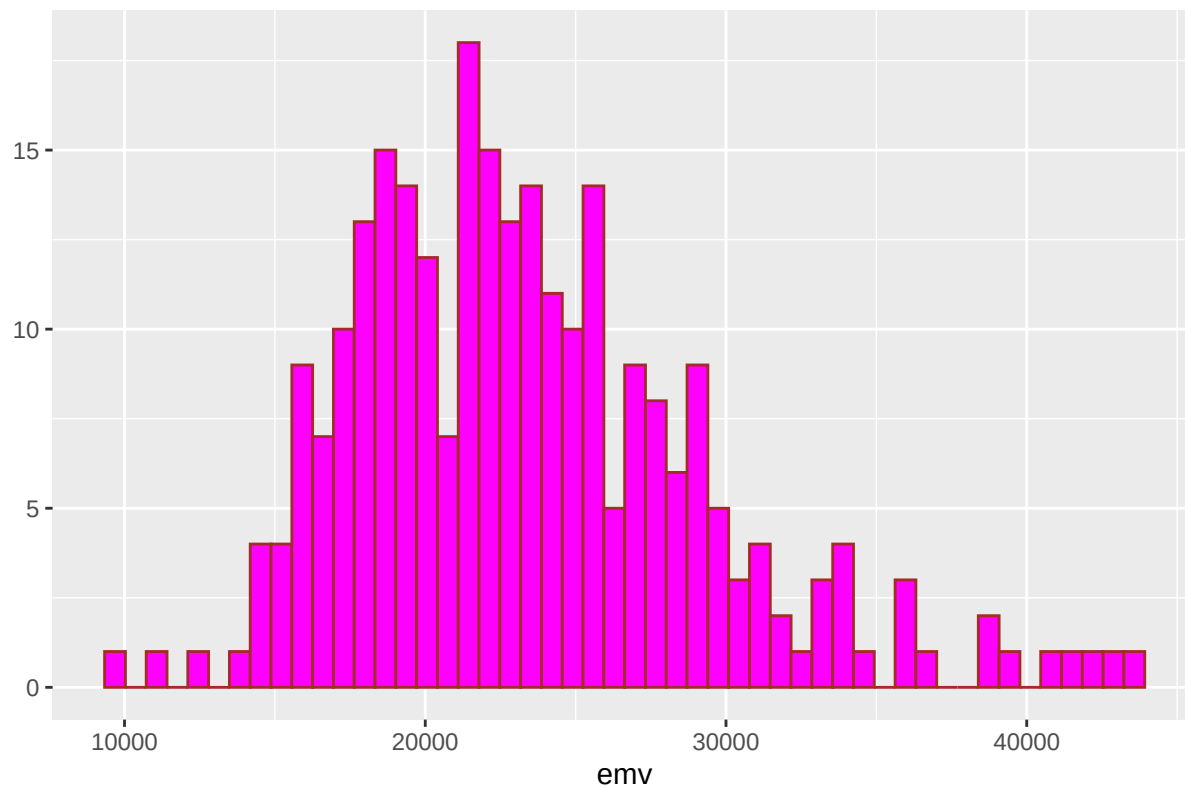
Q 2, part A

Display a histogram of the conditional distribution of emv given that luxury is in the interval $(.2, .3)$ in the cars dataset (from problem set 1).

```
library(DataAnalytics)
library(ggplot2)
data(mvehicles)
cars=mvehicles[mvehicles$bodytype != "Truck",]
# subset the conditional data of interest
cars_luxury.2.3=cars[cars$luxury >= .2 & cars$luxury <= .3,]
# plot the histogram
qplot(emv,
      data=cars_luxury.2.3,
      geom="histogram",
      bins=50,
      fill=I("magenta"),
      col=I("brown"),
      main = bquote("Histogram for emv conditional on " ~ 0.2 <= ~ "luxury" <= 0.3 ))
```

```
## Warning: `qplot()` was deprecated in ggplot2 3.4.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Histogram for emv conditional on $0.2 \leq \text{luxury} \leq 0.3$



Q 2, part B

Compute the mean of the conditional distribution in part A and compute a prediction interval that takes up 95% of the data (an interval that stretches from the .025 quantile (2.5 percentile) to the .975 (97.5 percentile)). Use the `quantile()` command.

```
mean(cars_luxury.2.3$emv)
```

```
## [1] 23376.5
```

```
quantile(cars_luxury.2.3$emv,probs=c(.025,.975)) ## prediction interval
```

```
##      2.5%      97.5%
```

```
## 14699.09 38632.34
```

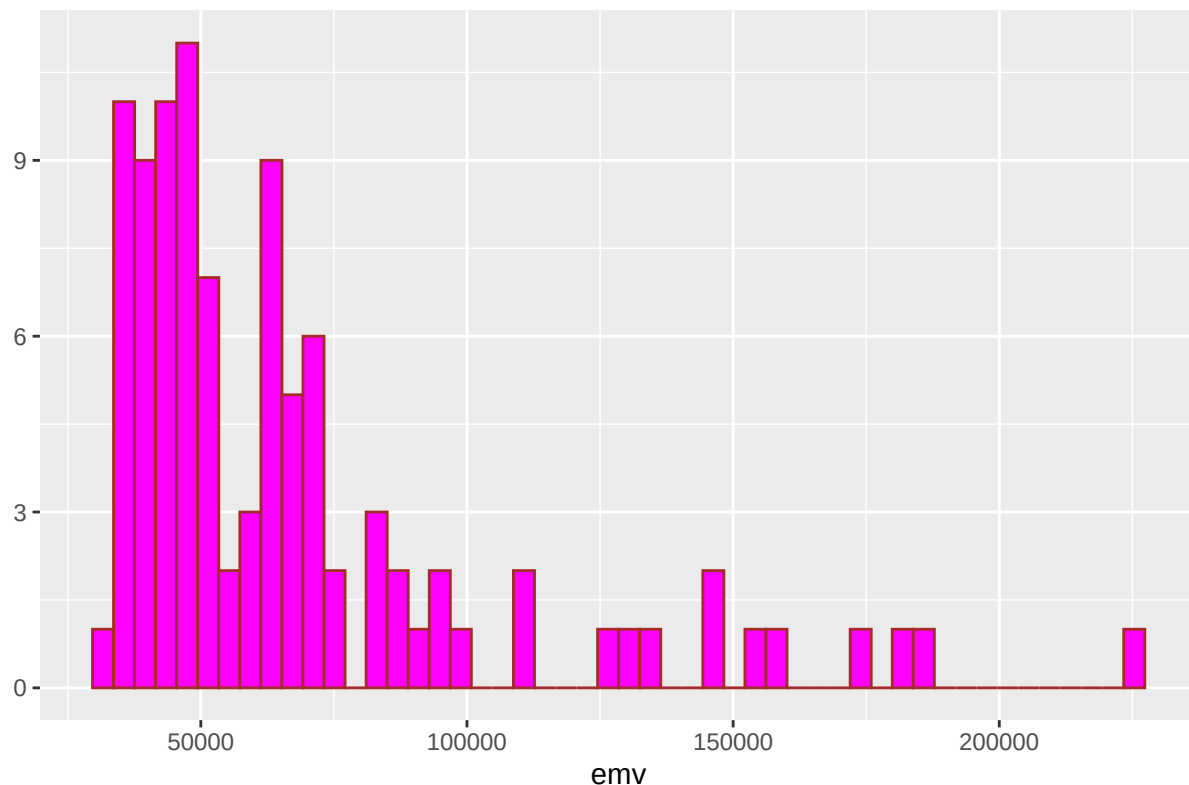
Q 2, part C

Repeat part A and B for a much higher level of luxury, namely the interval $(.7, .8)$. Describe the difference between these two conditional distributions.

```
# subset the conditional data of interest
cars_luxury.7.8=cars[cars$luxury >= .7 & cars$luxury <= .8,]
# plot the histogram
qplot(emv,
      data=cars_luxury.7.8,
      geom="histogram",
      bins=50,
      fill=I("magenta"),
      col=I("brown"),
```

```
main = bquote("Histogram for emv conditional on " ~ 0.7 <= ~ "luxury" <= 0.8 ))
```

Histogram for emv conditional on $0.7 \leq \text{luxury} \leq 0.8$



While the conditional distribution of estimated market value (emv) for vehicles low on luxury index is approximately normal, for those higher on the index, it is right skewed. This is to be expected as there will always be a small number of high market value vehicles.

```
mean(cars_luxury.7.8$emv)
```

```
## [1] 68101.87
```

```
quantile(cars_luxury.7.8$emv,probs=c(.025,.975)) ## prediction interval
```

```
##      2.5%      97.5%
```

```
## 34254.33 179179.22
```

The mean market value of high luxury vehicles is expectedly higher.

Q 2, part D

Explain why the results of part B and C show that luxury is probably (by itself) not sufficiently informative to give highly accurate predictions of emv.

The differences in the histograms and means show the market value varies with luxury index of the vehicle. However, the prediction intervals are very wide. Such high variation implies inaccuracies in predicted emv. There are likely other factors that we need to account for.

Question 3 : Optimal Pricing and Elasticities

Q 3, part A

Use the `detergent` dataset to determine the price elasticity of demand for 128 oz Tide. Compute the 90 percent confidence interval for this elasticity.

```
data(detergent) ## load the data
# regress log quantity on log price - the price coefficient represents elasticity
outlm=lm(log(q_tide128)~log(p_tide128),data=detergent)
lmSumm(outlm)

## Multiple Regression Analysis:
##      2 regressors(including intercept) and 14745 observations
##
## lm(formula = log(q_tide128) ~ log(p_tide128), data = detergent)
##
## Coefficients:
##              Estimate Std Error t value p value
## (Intercept)    13.300    0.13690   97.14      0
## log(p_tide128)  -4.412    0.06452  -68.38      0
## ---
## Standard Error of the Regression:  0.7358
## Multiple R-squared:  0.241  Adjusted R-squared:  0.241
## Overall F stat: 4675.84 on 1 and 14743 DF, pvalue= 0

# 90% CI for the regression estimates
confint(outlm,level=.90)

##              5 %      95 %
## (Intercept)  13.072601 13.522963
## log(p_tide128) -4.518186 -4.305912
```

Q 3, part B

*One simple rule of pricing is the “inverse elasticity” rule that the optimal gross margin should be equal to the reciprocal of the absolute value of the price elasticity, i.e. $\text{Gross Margin} = \frac{1}{|\text{elasticity}|}$. For example, suppose we estimate that the price elasticity is -2 (a 1 per cent increase in price will reduce sales (in units) by 2 per cent). Then the optimal gross margin is 50 percent. Suppose this retailer is earning a 25 per cent gross margin on 128 oz Tide. Use CI computed in part (a) and observe that whether this value is in the CI.

The elasticity consistent with a gross margin of 25% is $-4.0 \left(\frac{1}{|\text{elasticity}|} \right)$.

Since the CI interval does not include -4, it is not a reasonable estimate (non-significant) of epsilon.

Questions 4-5 explore the sampling properties of least squares

Question 4

a

Write your own function in R (using `function()`) to simulate from a simple regression model. This function should accept as inputs: b_0 (intercept), b_1 (slope), X (a vector of values), and σ (error standard deviation). You will need to use `rnorm()` to simulate errors from the normal distribution. The function should return a vector of Y values.

```
RegSim <- function(beta0, beta1, x, sigma) {
  # simulation of errors from normal distribution with mean 0 and user specified standar
  eps <- rnorm(n = length(x), mean = 0, sd = sigma)
```

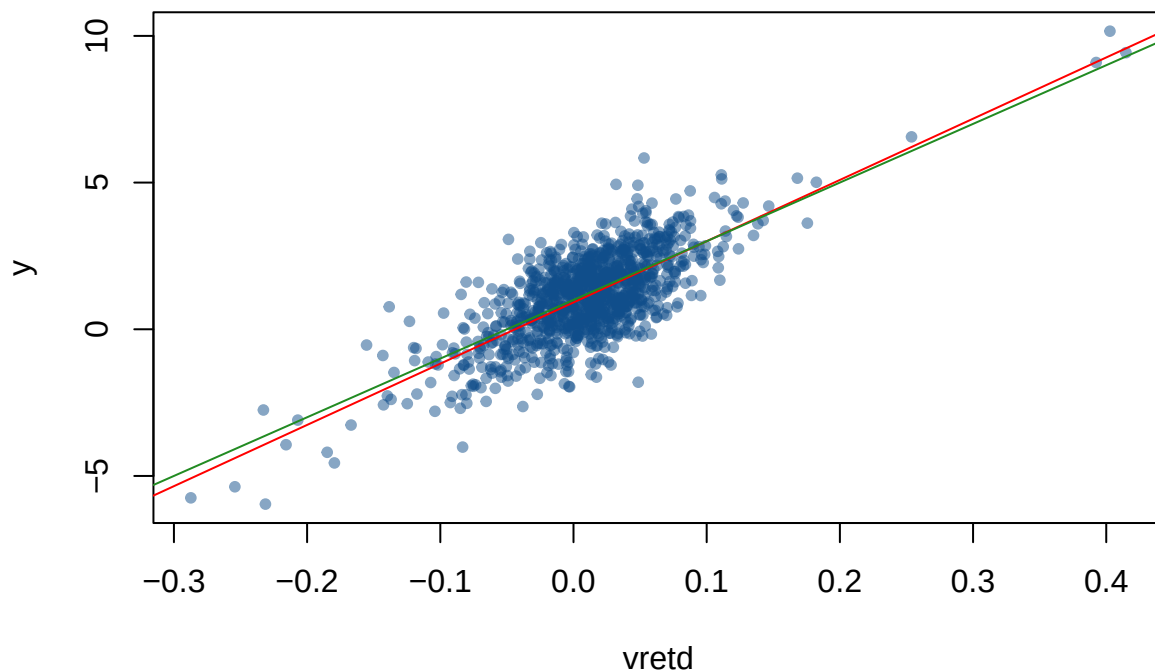
```
y <- beta0 + beta1*x + eps
}
```

b

Simulate Y values from your function and make a scatterplot of X versus simulated Y . When simulating, use the `vwretd` data from the `marketRf` dataset as the X vector, and choose $b_0 = 1$, $b_1 = 20$, and $\sigma = 1$. Then add the fitted regression line to the plot as well as the true conditional mean line (the function `abline()` may be helpful).

```
library(DataAnalytics)
data(marketRf)
set.seed(1991) ## for replication purposes
x <- marketRf$vwretd
y <- RegSim(beta0=1, beta1=20, x=x, sigma=1)
# scatter plot
plot(x, y, pch = 20, col=scales::alpha("dodgerblue4",0.5),
     xlab="vret", ylab="y", main="Simulated Regression")
# plot the estimated regression line on the scatter plot
abline(lm(y~x), col="red")
# plot the true regression line (or true conditional mean line)
abline(a=1, b=20, col="forestgreen")
```

Simulated Regression



Question 5

Assume $Y = \beta_0 + \beta_1 X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Let $\beta_0 = 2$, $\beta_1 = 0.6$, and $\sigma^2 = 2$. You can make X whatever you like, for example you could simulate X from a uniform distribution.

a

Use your R function from question 4 to simulate the sampling distribution of the slope. Use a sample size of $N = 300$ and calculate b_0 & b_1 for 10,000 samples. Plot a histogram of the sampling distribution of b_0 .

First, we generate some data, using $\beta_0 = 2$, $\beta_1 = .6$, and $\sigma^2 = 2$ using randomly generated X values with a sample size of 300. We estimate β_0 from this sample 10,000 times to generate the distribution of β_0 .

```
set.seed(1234)
R <- 10000 ## simulations
N <- 300 ## sample size
# define true parameter values
beta_0 <- 2
beta_1 <- 0.6
sigma <- sqrt(2)

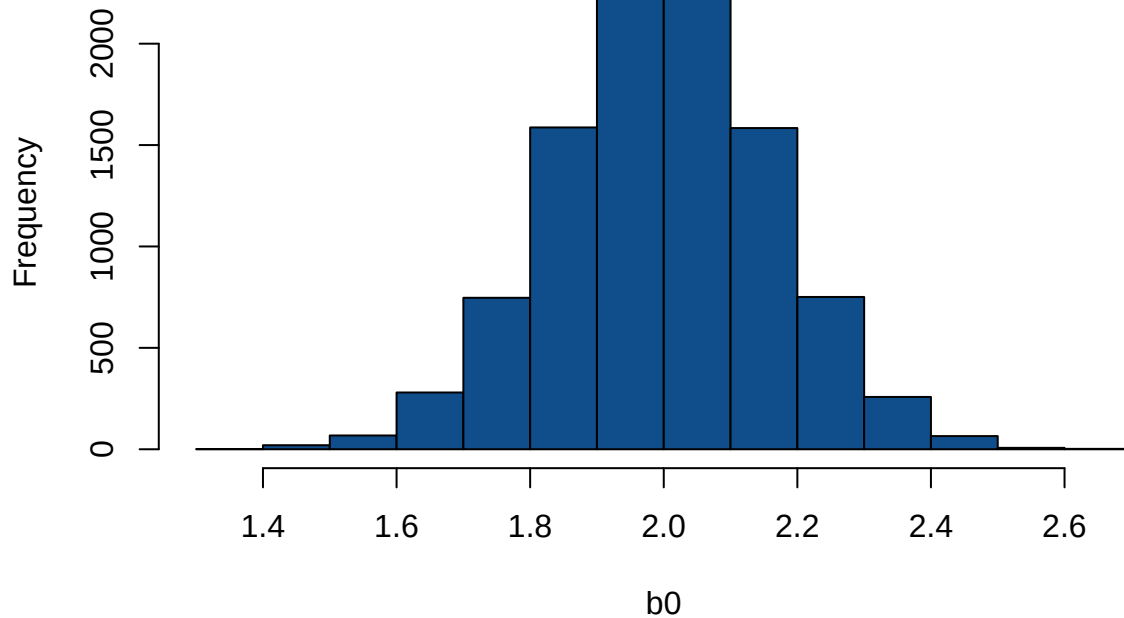
# simulate the values for X
X <- runif(N)

#vectors to store simulated intercepts
b0vec <- rep(NA, R)
b1vec <- rep(NA, R)

for (r in 1:R) {
  # generate Y values
  Y <- RegSim(beta0=beta_0, beta1=beta_1, x=X, sigma=sigma)
  # obtain and store least squares intercept and slope estimates
  b0vec[r] <- lm(Y~X)$coeff[1] ## intercept
  b1vec[r] <- lm(Y~X)$coeff[2] ## slope
}

#plot histogram
hist(b0vec, main = "Simulated Distribution of b0", xlab = "b0", col = "dodgerblue4")
```

Simulated Distribution of b0



b

Calculate the empirical value for $\mathbb{E}[b_1]$ from your simulation and provide the theoretical value for $\mathbb{E}[b_1]$. Compare the simulated and theoretical values.

```
# empirical value of E[b_1]
mean(b1vec)
```

```
## [1] 0.601349
```

The theoretical value for $E[b_1]$ is 0.6 (parameter value used to simulate the data). The simulated value is extremely close to the theoretical value. It shows that b_1 is an unbiased estimator of β_1 .

c

Calculate the empirical value for $\text{Var}(b_1)$ from your simulation and provide the theoretical value for $\text{Var}(b_1)$. Compare the simulated and theoretical values.

The theoretical value for $\text{Var}(b_1) = \frac{\sigma^2}{\sum_{i=1}^N (X_i - \bar{X})^2}$

```
# theoretical var(b1)
xbar <- mean(X)
SX <- sum((X-xbar)^2)
sigma^2/SX
```

```
## [1] 0.08480785
```

The empirical value is directly obtained from the vector of simulated b_1 values.

```
# empirical value of var(b1)
var(b1vec)
```

```
## [1] 0.08392873
```

We see that the variance from simulation is almost identical to its theoretical counterpart.

Question 6

Standard errors and p-values.

a

What is a standard error (of a sample statistic or an estimator)? How is a standard error different from a standard deviation?

As seen in the discussion during class, we are often interested in characterizing the variability associated with an estimator. Standard deviation is usually used to characterize the spread of a distribution. In the simulation exercise of the previous problem, we used exactly that.

Unfortunately, this standard deviation cannot be directly obtained from the data because it is defined over hypothetical repeated random sampling. In the simulation of previous question, we were able to compute it because we generated multiple data sets from the assumed data-generating process. However, in reality, we only ever have one data set and we almost always never know the true data generating process. We can still estimate the standard deviation of the sampling distribution (even though we can never see the sampling distribution) of an estimator from the observed data. The resulting estimate of the standard deviation of the sampling distribution is called standard error and it describes the estimated average degree to which an estimator deviates from its expected value.

In one line - The standard error of a sample statistic is the estimated standard deviation of the sampling distribution of that statistic

b

What is sampling error? How does the standard error capture sampling error?

We take a sample (preferably random) and use statistics from it to estimate a population parameter. The sample statistic is unlikely to be ever the same as the parameter it is estimating and the difference between the two is the sampling error. In the previous question, sampling error is the idea that each new sample leads to a new estimated value for b_0 and b_1 .

The standard error can be thought of as the estimated “size” of sampling error in the sense that a 95% confidence interval is ± 2 standard errors. Thus, the “margin of error” from sampling is basically two standard errors.

c

Your friend Steven is working as a data analyst and comes to you with some output from some statistical method that you’ve never heard of. Steven tells you that the output has both parameter estimates and standard errors. He then asks, “how do I interpret and use the standard errors?” What do you say to Steven to help him even though you don’t know what model is involved?

The standard errors represent the dispersion of the parameters in the output. It indicates the level of uncertainty around the estimate. An approximate 95% confidence interval would be the parameters plus or minus two standard errors.